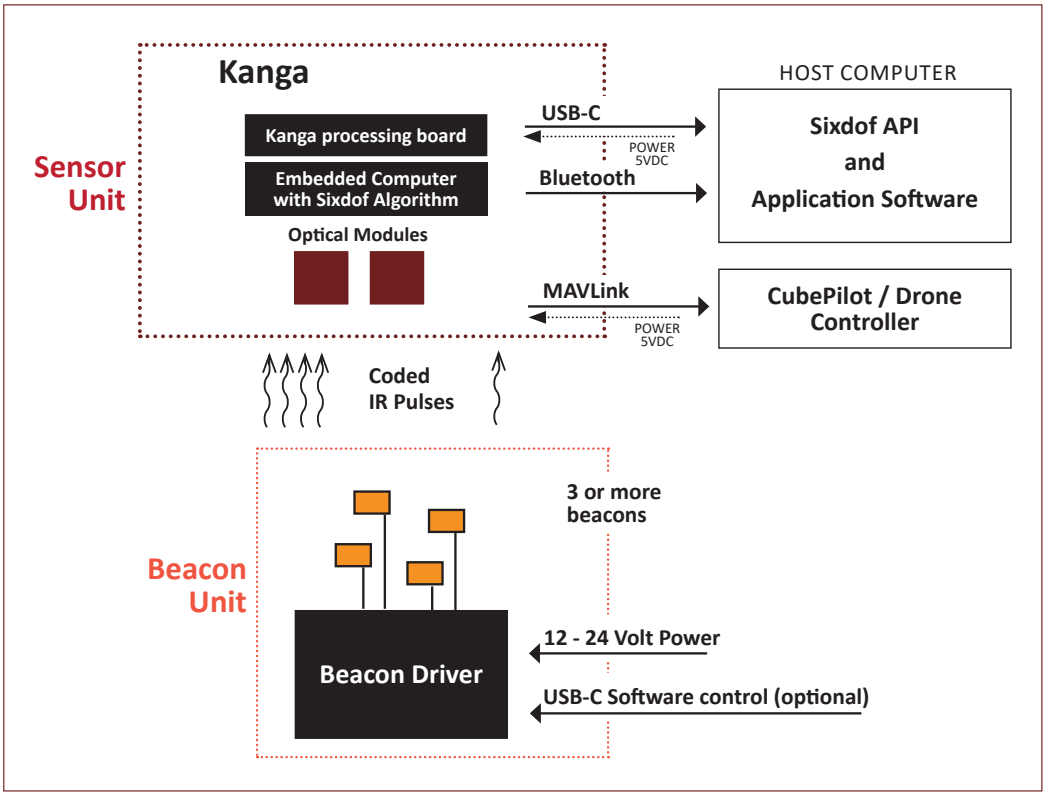
Sixdof Space
ULTRA FAST OPTICAL TRACKING

Kanga Sensor Data Sheet

The Kanga sensor's multiple proprietary optical modules detect the beacons in real time. The sensor then performs signal processing locally with an on-board FPGA. The compressed optical data is then sent to the onboard computer which calculates the sensor's full 6DOF position. The final estimate is then communicated to your flight controller real time over a MAVLink connection, or to your application code in real time over a USB or Bluetooth connection.

HIGHLIGHTS

- High-Resolution Drift-Free Optical Tracking
- Fully independent of GPS
- Operates under conditions of total darkness to full sunlight
- High toleration for strong winds and adverse weather conditions
- Includes Plug&Play MAVLink connection for CubePilot to manage drone landing
- Full 6DOF Pose (position and orientation) with cm-level accuracy
- Tracking up to 100 meters
- Simple installation and maintenance
- Option for USB or Bluetooth connection to host computer with C++ and Android API
- High Speed (updates every 2.5 ms)
- Compatible with Windows, Linux, Android, and Raspberry PI operating systems



BENEFITS

- Precision low-latency angular and 6DOF tracking
- Low computational requirements (no DSP/GPU)
- Continuous optical lock during high-rate movement
- Clearly defined API enables a range of use cases
- Software includes system integration examples

PERFORMANCE

Absolute Angular Accuracy
<0.5° over 120° FoV

Absolute Positional Accuracy
As low as 1 cm

(Dependent on distance and beacon baseline.)

Field-of-view
120° x 120°

Operating light levels
0 – 100,000 lux

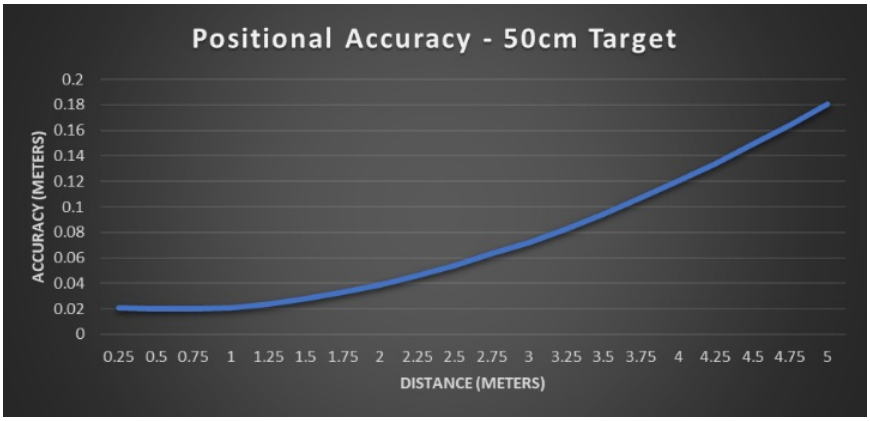
Interface Options
MAVLink, USB-C, Bluetooth

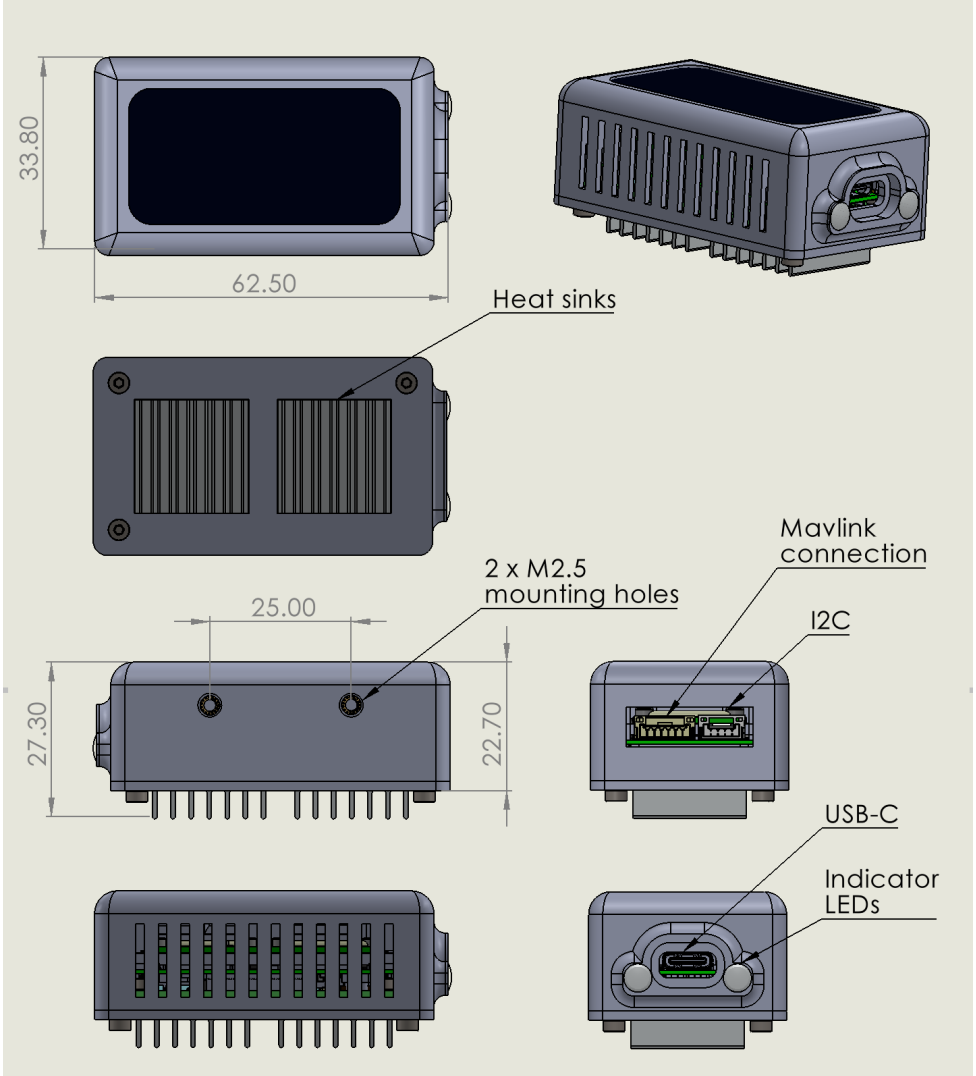
Power supply (sensor)
5V, 2.2 Watts
(Peak during
startup <3 Watts)

Operating temperature
-30°C to 50°C

Raw measurement frequency
Over 10,000 FPS

Capture latency
<3ms





PART NUMBERS

010-1011-001
Kanga Sensor with USB-C & MAVLink
(USB-C Connectivity requires software activation)

010-1011-002
Kanga Sensor with USB-C & Bluetooth
(Includes Water Resistant Housing)

Elevate your tracking and positioning technology with Sixdof's unmatched optical tracking precision to enable autonomous robotic and drone control.

For more details, visit us at www.sixdofspace.com